

Alex Bruno

Kenfack Anafack

// SUMMARY

Backend and machine-learning engineer with **8 years** across production Python systems and the teams that ship them. I design and deliver REST & GraphQL APIs, ETL pipelines and computer-vision & LLM/RAG products on Azure. As Founder & CEO of **AI Technologies for Africa** I have shipped award-winning, grant-funded AI products and secured client contracts with Expertise France and the Embassy of Spain; today I build core insurance systems at Equisoft in Montréal. I care about solid engineering that ships and creates real-world impact.

SPOKEN LANGUAGES French (native) · English (professional)

// TECHNICAL SKILLS

PROGRAMMING	Python, SQL, JavaScript, C++, C#
BACKEND & APIS	FastAPI, Django, Django REST, .NET, GraphQL, RabbitMQ, Nginx, Gunicorn
MACHINE LEARNING	PyTorch, OpenCV, Scikit-learn, OpenVINO, YOLOv8, ClearML
LLM & RAG	LangChain, OpenAI / Azure OpenAI, FAISS, RAG, AI Agents
DATA & ETL	Pandas, NumPy, Airflow, BeautifulSoup, Scrapy, Selenium, ElasticSearch
DATABASES	PostgreSQL, MySQL, MSSQL, Azure SQL
CLOUD & DEVOPS	Azure, Docker, Terraform, GitHub Actions, Azure DevOps, Azure App Service
3D & GEN-AI	3D Printing & Modeling, Slicing (Cura), Meta SAM, 3D Generative AI

// EXPERIENCE

Python Developer · Equisoft

feb 2025 to present

Montréal, QC

- Build and maintain a modular Python core package for life-insurance business rules, with cache, events, REST, GraphQL, DB, SMTP & SFTP modules consumed across product teams.
- Optimised asynchronous message exchange with RabbitMQ: dynamic queue management, custom consumers with failover handling and custom message providers.
- Introduced correlation and tracking IDs across logging and extended the Azure DevOps CI/CD pipeline, materially cutting debugging and release time.
- Write unit & integration tests and review pull requests to keep the shared core reliable; work day-to-day in an Agile/Scrum team on a mission-critical platform.

Founder & CEO · AI Technologies for Africa

jun 2022 to present

Yaoundé, Cameroon

- Founded and lead AITECAF, applying AI across the continent (crop-disease detection, computer vision, chatbots and IoT) alongside a digital agency practice; set technical direction and manage engineering and design teams.
- Architected and shipped flagship products: an AI fraud-risk audit tool, a RAG chatbot builder (OpenAI + LangChain) and a WhatsApp text/image generation bot, deployed on Azure.
- Secured client contracts with Expertise France (Auditrack and Prime, plus a maintenance contract) and the Embassy of Spain (Tryon).
- Secured ~3,500,000 FCFA in grants (Fonds Proto, TYSYP) and a £5,000 Afri-Plastics R&D grant; OpenCV AI Competition finalist and multiple IndabaX Cameroon wins.

Python / ML Developer • Nano Dev SARL

jan 2022 to jan 2024

Yaoundé, Cameroon

- Developed a PyTorch + OpenCV food-detection model paired with 3D reconstruction to estimate the volume and weight of detected items.
- Designed and tested an ETL pipeline collecting CSV/JSON/XML from Google Analytics, Google & Bing Ads, Facebook, Amazon S3 and ElasticSearch.
- Annotated datasets and built, tested and deployed machine-learning applications across client projects.
- Containerised models with Docker and exposed them through REST APIs for straightforward client integration.

Programmer Analyst • Pether Corporation

jan 2019 to jan 2022

Abuja & Douala

- Built insurance, hospital-management and claims systems with Angular front-ends and Django backends.
- Gathered and analysed client requirements, ran clarification workshops and designed customised solutions.
- Designed, integrated, tested and deployed business solutions, validated through user acceptance testing (UAT).
- Authored technical documentation and trained client staff on the delivered systems.

// SELECTED PROJECTS

AgriVision · crop-disease detection on cocoa plantations using AI and an autonomous drone; a smart camera spots diseased pods in flight and streams results to a multi-device decision dashboard.

stack · PyTorch, Fast-RCNN, YOLOv8, OpenCV, OAK-D, Drone / IoT

OpenCV AI Competition finalist (won 4 OAK-D cameras) · ~3.5M FCFA in grants · IndabaX Cameroon wins

Prime · AI audit assistant detecting fraud risk in the allocation of public contracts; an ETL pipeline ingests tender awards, ML scores each for fraud risk, and auditors are alerted automatically.

stack · Django, Machine Learning, Web Scraping, ETL, PostgreSQL, Twilio

built for the Audit Chamber of the Supreme Court of Cameroon; contracted via Expertise France

Peniang · multi-application platform connecting every stakeholder in the waste value chain (collectors, recyclers and producers) within a digital circular economy.

stack · Django REST, PostgreSQL, Azure Storage, OAuth2

Afri-Plastics Challenge Strand 3 semi-finalist (£5,000 R&D grant) · live in production

Sandbox · platform letting SMEs build intelligent website chatbots from just a URL, with retrieval-augmented generation over the crawled site, deployed in minutes.

stack · RAG, LangChain, OpenAI, FAISS, Django, AI Agents

Dona · full ChatGPT + DALL-E capabilities delivered through a WhatsApp chatbot for text and image generation, inside the chat people already use.

stack · WhatsApp API, LangChain, OpenAI, Azure OpenAI, FAISS

Tryon · AI virtual try-on platform: shoppers preview products on themselves from a single photo, powered by Azure AI inference and generative imaging.

stack · Django REST, Azure AI, OpenAI, Pillow, Azure Storage, OAuth2

delivered under contract with the Embassy of Spain

People Counter · smart-video IoT application detecting people in a zone and reporting live count, average dwell time and totals, running on Intel OpenVINO at the edge.

stack · OpenVINO, OpenCV, Python, Edge AI, IoT

// EDUCATION

BSc, Computer Science · University of Dschang, Cameroon

2014 to 2018

Junior Data Analyst, ParcoursTech · NPower Canada, Montréal

2025

// CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) · Microsoft

IBM Data Analyst Professional Certificate · IBM · Coursera · 11 courses

Intel Edge AI for IoT Developers (Nanodegree) · Intel / Udacity

AI Programming with Python (Nanodegree) · Udacity

Fundamentals of Computer Vision & Image Processing in Python · OpenCV University

Introduction to Machine Learning · Duke University · Coursera

Data Science Orientation · IBM · Coursera

// AWARDS & GRANTS

2023 **Winner, IndabaX Cameroon Ideathon** · pothole-detection AI built with PyTorch & YOLOv8.

2022 **Funded** · TYSYP, Ministry of Youth Affairs · 1,500,000 FCFA for the crop-disease AI & drone project.

2022 **Funded** · Fonds Proto (APME / MINPMEESA) · 2,000,000 FCFA prototyping grant for the same project.

2022 **Semi-finalist, Afri-Plastics Challenge (Strand 3)** · £5,000 R&D grant for the Peniang waste platform.

2021 **Finalist, OpenCV AI Competition (Phase 1)** · plant-disease detection with Fast-RCNN, PyTorch & OAK-D; won 4 OAK-D cameras.

// OPEN-SOURCE CONTRIBUTIONS

Ivy · unified machine-learning framework

scikit-image · image processing in Python

bugbug · machine learning for software engineering (Mozilla)